DaddiesTrip

Sit back, relax, let daddy handle the itinerary

Plan Trip

Itinerary

Expenses

UM Hackathon 2026 · UTM SugarDaddies

"Tell me about your dream trip — where, who, when, budget?"

e.g. 5 days in Tokyo for 4 people under RM5000...

Destinations

Trips Planned

Currencies

The Problem

Group travel planning is painfully fragmented across 4–5 apps

Search Flights

Juggling multiple airline sites and comparison tools

Find Hotels

Switching between Booking.com, Agoda, Airbnb…

Plan Itinerary

Manual day-by-day drafting in notes or spreadsheets

Split Costs

Manual currency calculations across group members

DaddiesTrip replaces all of them — one conversational interface, end-to-end.

Our Solution

1. Describe your trip in plain English (or voice)
2. AI extracts destination, dates, group size & budget
3. 4-agent pipeline generates a full day-by-day itinerary
4. Real flights & hotels with pricing — ready to book
5. Auto multi-currency expense split across all members

Sample Prompt

"5 days in Tokyo for 4 people under RM5000"

↓ **Generated Output**

- 5-day itinerary
- 3 flight options
- Hotel recommendations
- RM split in MYR & JPY

Key Features

Conversational Planning

Plain English prompt → structured day-by-day itinerary with activities & tips

Real-Time SSE Streaming

Progressive results streamed as each AI agent completes its task

Flight & Hotel Booking

Real airline pricing, times & direct links to Skyscanner / Google Flights

Multi-Currency Splitting

Live FX rates for 17+ currencies with deterministic offline fallback

Voice Input

Web Speech API — hands-free trip description with real-time transcription

PDF Export

One-click printable itinerary with flights, schedule, costs & hotel details

System Architecture

4-Agent Pipeline — strictly decoupled to minimise LLM hallucinations

01

Analyzer

LLM + Regex

Validates all 4 required fields; requests clarification if missing

→

02

Planner

LLM

Generates day-by-day itinerary with activities & transport

→

03

Booking

LLM

Enriches with real hotels, flights, POI costs & star ratings

→

04

Edge

Python Heuristics

Detects & corrects hallucinations — costs, routes, fields

Orchestrator streams SSE events at each stage — users see live progress

`progress → partial_itinerary → partial_flights → complete`

Tech Stack

Frontend	React 18 · Vite 5 · Tailwind CSS 3
Backend	FastAPI · Pydantic · Pandas
AI / LLM	Ilmu AI API (OpenAI-compatible) · Multi-agent Streaming
Currency	@fawazahmed0/currency-api · 17+ currencies · Offline fallback
Maps	Google Maps Embed API · Live POI iframe context
Deployment	Railway (backend Docker) · Vercel (frontend CDN)

Pipeline Deep Dive

1

Input Truncation

Prompts > 1,500 words truncated — prevents buffer overflow

2

Analyzer

Validates 4 required fields; clarification event if missing

3

Planner → SSE

Day-by-day skeleton streamed immediately as partial_itinerary

4

Booking → SSE

Real-world enrichment; partial_flights event emitted

5

Budget Calculation

Pure Python — aggregates costs, compares budget, saving tips

6

Currency Conversion

Live FX fetched; offline fallback for 17+ currencies

7

Edge Validation

Fixes repeated costs, invalid routes, missing fields

8

Final Emission

Validated complete event — full trip object sent to frontend

Deployment Architecture

Split deployment — optimised for long-running LLM pipelines (30–90s)

FRONTEND

Vercel

React 18 + Vite static build

- CDN-served, global edge network
- Automatic HTTPS
- Edge rewrites for API proxy
- Zero cold-start for static assets

BACKEND

Railway

Docker container (FastAPI)

- No hard timeout limits
- Supports 30–90s LLM pipelines
- Persistent Docker containers
- Auto-deploy from GitHub

⚠ Vercel Serverless Functions timeout at 10–60s — Railway runs with no hard limit, making it the right fit for LLM pipelines.

API Reference

Endpoints

Method	Endpoint	Description
POST	`/api/plan-trip-stream`	Primary pipeline trigger — streams SSE results
POST	`/api/amend-item`	Amend a hotel, food, or activity item
POST	`/api/settle`	Simulate group ledger card payment
GET	`/api/health`	Service health check

SSE Event Types

`progress`	`partial_itinerary`	`partial_flights`	`complete`	`error`
Pipeline stage begins	Planner complete	Booking complete	Full trip emitted	Fatal failure

Testing

PyTest suite covering end-to-end pipeline, payment rejection, and input safety

TC-01

End-to-End Pipeline

Validates full SSE pipeline output schema — itinerary, flights, budget & split — from a valid prompt

TC-02

Deterministic Ledger Rejection

Verifies invalid card numbers are correctly rejected by the payment settlement endpoint

AI-01

Buffer Overflow Protection

Ensures oversized prompts are truncated to 1,500 words before reaching the LLM

```
$ python -m pytest backend/tests/test_agents.py -v
```

Getting Started

1. **Clone the repo**

```
git clone github.com/SkyLee310/UMHackathon-DaddiesTrip
```

2. **Configure .env**

```
cp .env.example .env  →  add your API key
```

3. **Start backend**

```
pip install -r requirements.txt
uvicorn backend.main:app --reload
```

4. **Start frontend**

```
cd frontend && npm install && npm run dev
```

Environment Variables

`Z_AI_API_KEY`

Required — LLM service key

`Z_AI_BASE_URL`

LLM API base URL

`Z_AI_MODEL`

ilmu-glm-5.1

`VITE_API_BASE_URL`

Backend URL for frontend

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SugarDaddies: We take care of everything. Especially you.

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